FERRITIC/AUSTENITIC TRANSITION WELD + NDE PROPOSAL

A dissimilar transition weld between ferritic steel and austenitic stainless steel is considered for a **high pressure corrosive system**.

WELDING

MATERIAL PROPERTIES

Ferritic: needs to withstand high pressure forces (strength) and absorb lots of energy before fracture (toughness)

Austenite: needs to withstand highly corrosive environments (corrosion resistance)

Weld filler: should be at least equal in strength and corrosion properties to the poorest component in the joint

MATERIAL SELECTION

Weld is between a **low carbon alloy steel (SA508)** and an **austenitic stainless steel (305)** using an austenitic **weld filler (309)**

Material	C	Si	Mn	Ni	Cr	Mo	Nb	Cr Equiv	Ni Equiv
SA 508	0.19	0.1	0.25	3.4	1.6	0.45	0.01	2	9
Type 305	0.05	0.25	1.0	10	19.0	0.05	0.05	19	12
Type 309	0.01	0.65	1.0	12	25.0	0.25	0.01	26	13

CHEMICAL COMPOSITION

Carbon: Precipitation and solid solution strengthening.

Silicon: Used for steelmaking purposes; killing the steel and removing oxygen. i.e improves resistance to oxidation. (important for high integrity)

Manganese: Improves strength and hardness

Chromium & Nickel: Primarily to drive corrosion resistance, but nickel can also improve toughness and hardenability.

Molybdenum: Enhances creep resistance

Niobium: Grain refinement and precipitation hardening

NICKEL AND CHROMIUM EQUIVALENTS

The **nickel equivalent** and **chromium equivalent** differ from the nickel content and chromium content, respectively.

$$\text{Cr Equiv.} = Cr + Mo + 1.5Si + 0.5Ni \qquad \text{Ni Equiv.} = Ni + 30C + 0.5Mn$$

Higher chromium equivalent will promote ferritic microstructure as its composed of **ferritic stabilisers**, likewise higher nickel equivalent will promote austenitic structure as its composed of **austenite stabilisers**.

WELDING CONSIDERATIONS

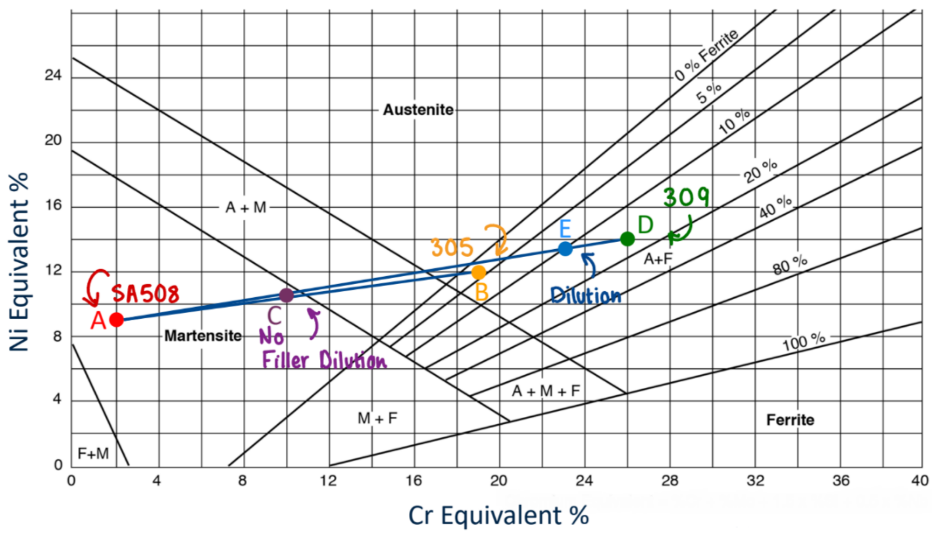
Dilution control: the resultant chemical composition of the fusion zone is related to the degree of mixing between the filler metal and the two parent metals.

Sensitisation: where carbides form from chromium, and essentially deplete the chromium content.

Unwanted phases: if metal is unstable, delta ferrite and sigma ferrite phases can form and degrade to brittle intermetallic phases, becoming susceptible to corrosion and fracture.

Ferrite Number (FN): A low ferrite number suggests larger proportion of austenite and hence better corrosion resistance, while a high ferrite number infers higher cracking resistance. Ideal FN for dilution of weld filler in this case = 10, which gives **good balance of crack resistance and corrosion properties**.

SCHAEFFLER DIAGRAM



Point A: SA508 Ferritic

Point B: 305 Austenite

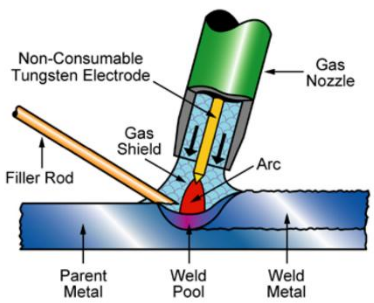
Point C: Welded area if no filler weld was used

Point D: 309 Filler

Point E: Target composition (FN=10) of weld controlled by dilution factor

Position (and hence composition) of E also depends on the size of the weld. A wider weld means a larger dilution factor and hence larger displacement on Schaeffler diagram. See below for more info on the geometric weld design:

WELDING PROCESS



The weld volume in this context is very small with **steep fusion angles of about 4 degre**es, resulting in very tight join gaps. Hence there isn't much of a dilution factor.

TIG (Tungsten Inert Gas) performs well in this kind of weld design as it is well suited for narrow joints and steep angles with fine control of heat input. Inert gas also prevents oxidation

NON-DESTRUCTIVE EVALUATION (NDE)

NDE is comprised of seven main areas, two of which (highlighted) are the main focus for this study: **Flaw detection and evaluation and Location determination and evaluation.**

NDE METHODS: RADIOGRAPHIC VS ULTRASONIC TESTING

NDE method needs to be a volumetric inspection method that is able to both locate and size the defects; the choice is between radiographic testing (RT) and ultrasonic testing (UT).

UT is very versatile it is able to size and locate defects, whereas RT don't offer the same resolution in depth due to the nature of radiography and hence locating defects is more difficult using RT.

TYPES OF ULTRASONIC TESTING:

Longitudinal waves: produced by voltage applied to a piezo-electric transducer.
Transverse (shear) waves: require more acoustically solid material for effective propagation (not effective in liquids and gases).

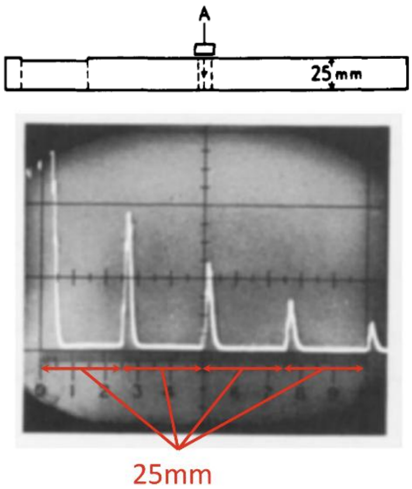
Longitudinal waves are approx. twice the speed of shear waves, hence wavelength (resolution) of transverse waves is roughly half, making it more sensitive to smaller defects.

However, **propagation of transverse waves is more difficult through anisotropic structure of weld**. So ultimately, sizing tolerance is hindered because longitudinal waves (which suffer from worse resolution due to larger wavelength) are more effective.

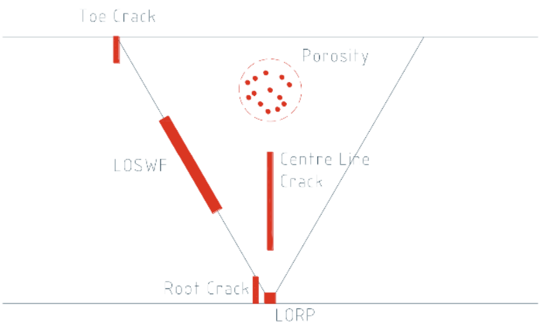
HOW DOES UT WORK:

Signals representing interface reflection are produced: Initial bang followed my evenly spaced repeating backwalls (intervals depending on component thickness) of decreasing amplitude.

If there is a defect, the waves will reflect in between these fixed intervals. This will appear as a smaller sharper **new echo (peak) before** the first backwall echo depending on the defect size/orientation.



DEFECT TYPES & DETECTION METHOD:



Toe Crack: two techniques could be used:
- ➔ Transverse waves from a 45° probe
- ➔ Creep waves (78° longitudinal surface waves)

LOSWF: best response given by 60° transverse probe which will strike defect at 90°
LORP: 45°, 60° or 45° transverse probes from either side of centreline

Root Crack: similar to LORP, but will not produce defined rise/fall signals and instead will produce multiple peaks and troughs.
Porosity: Difficult with UT as signals produced would be very weak, RT can be more effective
Centreline Crack: uses a transmitting shear probe and a receiving shear probe in a tandem setup.

Defect Type	Detection Method
Toe Crack	UT; transverse waves at 45°, or creep waves
LOSWF	UT; transverse waves at 60°
LORP	UT; transverse probes preferred at 45°
Root Crack	UT; transverse probes preferred at 45°
Porosity	Difficult with UT; RT can be more effective
Centreline Crack	UT; using two transverse probes

MISCELLANOUS – QUESTIONS & CONSIDERATIONS

How do we filter out more esoteric alloys in material selection?
Exotic materials can have excellent properties which are far in excess of what's required -- most popular ones tend to be the cheaper ones, which we can manufacture in bulk. They need to be produce to a large scale and sufficient volume, hence cost becomes a factor that cannot be ignored

Material index considerations
We need to maximise both fracture toughness and yield strength, not necessarily a ratio between them. Density is not an important factor for this context.

Further to why TIG? Why not electron beam welding?
TIG allows for automated mechanised welding which gives consistency and fine precision control.
For narrow gap welding applications (very small weld volume) and therefore steep preparation angles, precision needed as the toe can be difficult to reach.
TIG is preferred as Argon used as the shrouding inert gas, which produces a narrow, focused arc, which helps in achieving deeper penetration and a cleaner weld profile.
EBM, especially on large vessels cannot be used because we cannot fit large structures into a vacuum chamber.

TYRE ROAD WEAR PARTICLES (TRWP)

BACKGROUND

Design optimal passenger car tyre tread compound that reduces TRWP emissions, **maintains/improves key tyre performance** metrics such as wear resistance, rolling resistance (fuel economy), and wet grip, and finally minimises the environmental/health impacts of materials used.

SMEAR WEAR AND ABRASIVE WEAR

ABRASIVE WEAR

Crack formations develop on the macro scale at evenly spaced intervals. High frictional energy hence road surfaces act like cutting tools generating more debris.
Thermo-mechanical degradation: Accelerated particle detachment and abrasive (and fatigue) wear

SMEAR WEAR

Smaller micro-crack formations occur in much higher frequency. Generates 1/10th the volume abrasion. Low frictional energy causing small localised plastic deformation.
Tribo-chemical degradation: an oxidised sticky layer protects the bulk rubber surface from excessive crack initiation and propagation

GOAL

Although toxicity of smear wear is unknown relative to abrasive wear, smear wear produces 11/10th the volume, making smear wear preferred over abrasive wear.

A design implication of this is to increase the frictional energy threshold (through high toughness compounds) to ensure driving conditions stay below the threshold and promote smear wear.

TARGET OF TRWP DESIGN

Reduce impact of materials harmful to health/environment, normally associated with additives such as 6PPD which reacts with oxygen and forms harmful byproducts.
Reduce the wear rate to mitigate both volu,e and size impact of TRWP; design high toughness compound preferring smearing wear
Keep the better performance of rolling resistance and wet grip; solve trade off between them

BASE POLYMERS

Natural Rubber (NR); superior wear resistance, fair wet traction, superior energy loss (rolling resistance is low), excellent tear resistance. (highlight; low rolling resistance)
Styrene Butadiene Rubber (SBR); superior wear resistance, superior wet traction, fair energy loss, inferior tear resistance (highlight; good grip)

FILLERS

High surface area silica and limited high-structure carbon black used as fillers. Silica improves wet grip and reduces rolling resistance, carbon black contributes to wear resistance. Coupling agents used to bond silica to rubber matrix.

ADDITIVES:

TDAE oil; environmentally friendly processing oil. Alternative anti-oxidants and zinc oxide to replace toxic components like 6PPD

VISCOELASTIC RESPONSE & TYRE PERFORMANCE

Viscoelastic behaviour of the tread should be such that there is high energy loss for efficient wet traction, and low energy loss for rolling resistances in interest of fuel economy; high surface area silica fillers help achieve this as discovered by Michelin.

For high frequency cycling, viscoelastic properties are measured using time temperature superposition; low temperature behaviour represents high frequency behaviour.

NANOSTRUCTURE BLEND

Trade-off problem between good wear resistance and enhanced wet grip/toughness lends to <u>a bi-continuous polymer blend solution</u>:
→**Filled SBR**; for high grip (high toughness and higher hysteresis)
High loading filler gives better wear resistance and saturation of energy loss (hysteresis) / wet grip
→**Unfilled NR**; some good aspects of abrasion resistance with very low rolling resistance (lower energy loss and higher toughness)

This mitigates the compromise between wear resistance and rolling resistance and incorporates both.

CHAIN END MODIFICATION

In general, fillers are needed to enhance performance. However, in a bi-continuous phase such as this one, SBR needs fillers to enhance wet traction whereas NR should avoid fillers to preserve lower hysteresis.

To mitigate this, **SBR is functionalised** so that it contains end groups which are more attractive and therefore the fillers stick to the SBR phase, and <u>not</u> the NR phase.

MISCELLANOUS – QUESTIONS & CONSIDERATIONS:

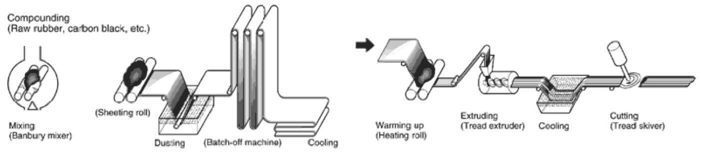
OPTIMISATION OF FILLERS

FILLER SIZE: Fine particle size (<100nm) improve mechanical properties and reduce wear by enabling better dispersion in polymer filler bonding (due to higher surface area)

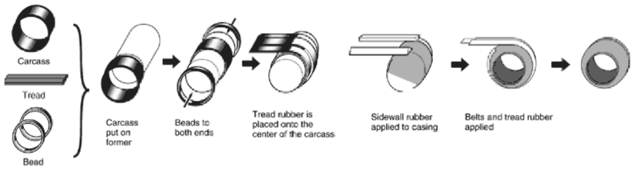
FILLER SHAPE: Affects toughness

SURFACE CHEMISTRY:

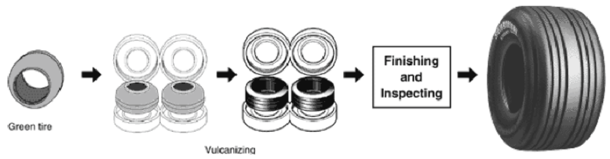
Mixing and extruding process



Assembling process



Vulcanizing process



INTERBODY CAGES FOR SPINAL FUSION

Design an **interbody spinal cage** to fuse the two vertebrae (spinal fusion) to treat back pain. The focus is to create a design that has an adaptive modulus, and allows the bone formed to bear an increasing load over time.

DESIGN CONSIDERATIONS

Patient safety: the ability to bear load until it is certain that new bone will have formed. Involves aligning the temporal adaptive modulus with bone formation.

Design features that **drive bone formation**; macro/micro porosity shapes and surfaces, bioactive coatings

Ability to assess bone formation (**radiolucency**)

DESIGN BACKGROUND

Titanium is used as a high strength osteoconductive material. But as per Wolff's Law, high stiffness can make the cage subject to stress shielding. Titanium is also radiopaque (illuminated by x-ray, hindering visibility of bone growth and creating artefacts).

PEEK is a radiolucent polymer which can have a modulus readily comparable to that of bone, mitigating stress shielding. But it is also hydrophobic and bioinert, can form fibrous tissue through foreign body response, leading to poor osteointegration.

Other polymers are biocompatible and degradable but suffer from poor load bearing. It can be difficult to maintain strength overtime due to degradation and loss of molecular weight- deformation is also a concern.

CONTRIBUTION TO BONE FORMATION

Porous structure: allows nutrients, blood vessels and bone cells to move through the cage, allowing bone on-growth.

Further to that: Biologically, osteoclasts come in and reabsorb bone, creating **concave surfaces** allowing cells to come and lay new bone to replace it. By mimicking these concave surfaces, we can potentially encourage cells into move into these biomimetic cues, where length scale (submicron, increased surface area) come to play. Using the same logic, **submicron surface features** can also promote bone growth. This process relies on osteoinduction as opposed to osteoconduction.

Idea is to provide a macro pore for angiogenic growth, and a smart concavity for microenvironment; enabling osteoinduction.

Bioactive coatings such as rhBMP-2 and HA (hydroxyapatite) can promote bone growth on bio-inert surfaces such as PEEK.

RADIOLUCENCY

Radiopaque materials (titanium etc) offer the stiffness and strength required but illuminate under x-ray and hinder the visibility of bone formation. Radiolucent materials (PEEK, polymers etc) allow clear imaging but lack sufficient biocompatibility for bone integration.

Idea is to use **radiolucent resorbable cores** with **porous radiopaque structural supports**; the design becomes more radiolucent as resorbable parts degrade (where bone monitoring is most vital).

PATIENT SAFETY

Anterior column bears about 60% of the spine weight, whereas the interior column bears 40%.

Apophseal ring, which bears most of the weight: very strong cortical bone. The centre of the vertebral body contains the softer, cancellous bone. Implications this has on our cage design is that we need to **use two materials to overcome the limitations of using just one material.**

This means we use a stiff material (Ti, Si3N4, Coated PEEK) as the permanent outer layer, and an innovative resorbable material (Mg Alloy) for the inner layer. Resorbable centre is mostly made from metal to allow it to be sterilised.

Challenges with **magnesium mean that will degrade very quickly** posing structural risk to forming bone and potentially release gases. We can use a coating (chitosan) to temper and slow the degradation down.

We use porosity, and innovative lattice structures to **offset modulus issues** with harder materials (like Ti) to mitigate stress shielding.

To highlight safety, we can also use **antimicrobial coatings** such as **silver** to prevent biofilm infections. Note: the negative impact of silver on bone formation can be negligible as long as other design features are included.

SOLUTION

3D printed Ti outer to sit on the Apophseal Ring: Open, porous structure with concave pores and submicron surface features.

A slot-in resorbable inner made mainly from a Mg alloy to allow personalisation at point of use (fast, medium and slow depending on Mg content) likely coated (chitosan).

In the very centre, a bone graft chamber for point of use addition of bone graft substitutes (rhBMP-2 coatings can be used to promote osteoinduction)

MANUFACTURING

Recent advances in 3D printing allows customisable designs with tailored and optimised geometry, increased porosity, and potential of using other materials.

3D printing processes can be optimised for osteoconduction, allowing surface modifications for bone on growth, and mimicking highly porous structure and optimum pore size of cancellous bone. Overall enabling biomimetic design.

3D-printed titanium spinal cages are typically manufactured using Selective Laser Sintering (SLS) After printing, post-processing includes powder removal, heat treatment to relieve stress, and surface treatments to enhance bone integration. Implants then undergo quality control through mechanical and dimensional testing. Finally, they are sterilized using gamma radiation.

MISCELLANOUS – QUESTIONS & CONSIDERATIONS:

Osteoconduction vs Osteoinduction

Osteoconduction: passive scaffold allowing stem cells to migrate across, where there is already a continuous connection to bone
Osteoinduction: process of bone growing away from other bone (without presence of surrounding bone) and sending them down a commitment pathway towards osteoblasts once they have grown. E.g BMP can be grown in muscle pouches, despite there being no surrounding bones.
Nanoscale Ti as coating for PEEK

Comments on carbon fibre coated PEEK:

Between **PEEK** and **Titanium**, titanium is too stiff, PEEK not stiff enough: use carbon fibre coating?
PEEK's issue is not that it's weak- cell growth is more difficult with PEEK as it's difficult to create the required pore size and shapes, relative to titanium. Market is leaning more towards titanium
Carbon fibres may not always be fully embedded which poses a risk as bits of fibre break off and contaminate the body.